LIM-2.D.3 The concept of a limit can be extended to include limits at infinity.

LIM-2.D.4 Limits at infinity describe end behavior.

LIM-2.D.5 Relative magnitudes of functions and their rates of change can be compared using limits.

1-15 Limits at Infinity and Horizontal Asymptotes

Get Ready...

Complete the table below to find $\lim_{x\to\infty} \frac{3x^2-2}{x^2+4x+1} = 3$
(use your calculator)

x	1	10	100	1000	10000	100000
$\frac{3x^2-2}{x^2+4x+1}$	0.166666666	2.113475177	2.884145755	2.9880428405	2.998800429	2.9998800004

Consider the following limit statements:

$$\lim_{x \to 3} f(x) = +\infty \qquad \lim_{x \to \infty} f(x) = 3$$

What do each of the limit statements say about the graph of $f(x)$?

Statement #1: There is a vertical asymptote at x = 3

Statement #2: There is a horizontal asymptote at y = 3

Previously, we looked at infinite limits and how they are connected to vertical asymptotes. Now, we will look at limits at infinity and their connection to horizontal asymptotes.

<u>Example 1:</u> Find $\lim_{x\to\infty} \frac{3x^2-5x}{2x^2+7}$ (**HINT**: When working with limits as x approaches infinity $\lim_{x\to\infty} f(x)$, **DON'T SWEAT THE SMALL STUFF!** We only need to compare the terms of greatest magnitude in the numerator and denominator!)

$$\lim_{x\to\infty} \frac{3x^2-5x}{2x^2+7} = \lim_{x\to\infty} \frac{3x^2}{2x^2}$$

$$= \lim_{x\to\infty} \frac{3}{2} = \frac{3}{2}$$

When taking the limit of a fractional expression to $\pm\infty$:
1.) If the denominator grows larger faster than the numerator, the limit is 0
2.) If the numerator grows larger faster than the denominator, the limit is $\pm\infty$
3.) If the numerator and denominator grow "equally," then you need to investigate further…the limit probably approaches a value.

Find the following limits, if possible.

<u>Example 2</u>: $\lim_{x\to\infty} \frac{x^2+4x-6}{5x^2+9} = \lim_{x\to\infty} \frac{x^2}{5x^2} = \frac{1}{5}$

Find the following limits, if possible.

Example 3: $\lim_{x\to\infty} \frac{(4-3x)(2x+1)}{3x^2-14} = \lim_{x\to\infty} \frac{-6x^2}{3x^2} = \frac{-6}{3} = -2$

Find the following limits, if possible.

Example 4: $\lim_{x\to\infty} \frac{4x^2+x-1}{x^3-2} = \lim_{x\to\infty} \frac{4x^2}{x^3} = \lim_{x\to\infty} \frac{4}{x} = 0$

Find the following limits, if possible.

Example 5: $\lim_{x\to\infty} \frac{2x^2-x-1}{3x-4} = \lim_{x\to\infty} \frac{2x^2}{3x} = \lim_{x\to\infty} \frac{2x}{3}$

$= +\infty$

Find the following limits, if possible.

Example 6: $\lim_{x\to\infty} \frac{(3x-7)^2}{12-3x-5x^2}$ $= \lim_{x\to\infty} \frac{9x^2}{-5x^2} = \frac{-9}{5}$

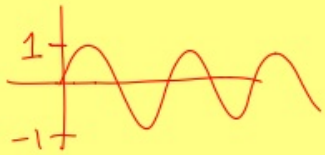
Find the following limits, if possible.

Example 7: $\lim_{x\to\infty} \frac{\sin(x)}{x}$

$-1 \le k \le 1$

$= \lim_{x\to\infty} \frac{|k|}{x}$

$= 0$



Find the following limits, if possible.

Example 8: $\lim_{x\to-\infty} \frac{2x^2-x-1}{3x-4} = \lim_{x\to-\infty} \frac{2x^2}{3x} = \lim_{x\to-\infty} \frac{2x}{3}$

$= -\infty$

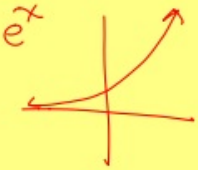
Find the following limits, if possible.

Example 9: $\lim_{x\to-\infty} \frac{(4x+1)^2}{8x^2+11x+19} = \lim_{x\to-\infty} \frac{16x^2}{8x^2} = \frac{16}{8} = 2$

Find the following limits, if possible.

Example 10: $\lim_{x\to\infty} \frac{x^{99}+5}{e^x+3x}$ $= \lim_{x\to\infty} \frac{x^{99}}{e^x} = 0$

e^x

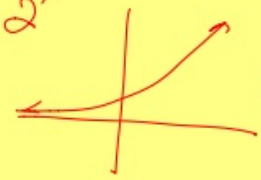


Exp. grow faster than powers

Find the following limits, if possible.

Example 11: $\lim_{x\to-\infty} \frac{4+2^x}{2^x-3}$ $= \lim_{x\to+\infty} \frac{4+2^{-x}}{2^{-x}-3} = \lim_{x\to\infty} \frac{4+\frac{1}{2^x}}{\frac{1}{2^x}-3}$

2^x



$2^x \to 0$ as $x \to -\infty$

$= -\frac{4}{3}$

Find the following limits, if possible.

Example 12: $\lim_{x\to\infty} \frac{\sqrt{4x^2-3}}{7x+5} = \lim_{x\to\infty} \frac{\sqrt{4x^2}}{7x}$

$= \lim_{x\to+\infty} \frac{2x}{7x} = \frac{2}{7}$

Find the following limits, if possible.

Example 13: $\lim_{x\to-\infty} \frac{\sqrt{4x^2-3}}{7x+5}$ $= \lim_{x\to-\infty} \frac{\sqrt{4x^2}}{7x}$

$= \lim_{x\to-\infty} \frac{2|x|}{7x} = \lim_{x\to-\infty} \frac{-2x}{7x} = \frac{-2}{7}$

Definition of Horizontal Asymptotes	$y = \mathrm{L}$ is a horizontal asymptote to the function $f(x)$ if… $\lim_{x \to +\infty} f(x) = \mathrm{L}$ or $\lim_{x \to -\infty} f(x) = \mathrm{L}$ **Note**: When finding horizontal asymptotes, we must consider **BOTH** limits…there could be different horizontal asymptotes!

Example 14: Find any horizontal asymptotes for the following functions.

a. $f(x) = \dfrac{x^2 - 3x + 4}{3x^2 + 5}$

HA: $y = \frac{1}{3}$

$$y = \lim_{x \to \infty} \frac{x^2 - 3x + 4}{3x^2 + 5} = \lim_{x \to \infty} \frac{x^2}{3x^2} = \frac{1}{3}$$

b. $g(x) = \dfrac{3x - 1}{x^2 + 2x + 7}$

HA: $y = 0$

$$\lim_{x \to \infty} \frac{3x - 1}{x^2 + 2x + 7} = \lim_{x \to \infty} \frac{3x}{x^2} = 0$$

Example 14: Find any horizontal asymptotes for the following functions.

c. $h(x) = \dfrac{3x-5}{\sqrt{5x^2+3}}$

HA: $y = \frac{3}{\sqrt{5}}$, $y = \frac{-3}{\sqrt{5}}$

$$\lim_{x\to+\infty} \frac{3x-5}{\sqrt{5x^2+3}} = \lim_{x\to\infty} \frac{3x}{\sqrt{5x^2}} = \lim_{x\to\infty} \frac{3x}{x\sqrt{5}} = \frac{3}{\sqrt{5}}$$

$$\lim_{x\to-\infty} \frac{3x-5}{\sqrt{5x^2+3}} = \lim_{x\to-\infty} \frac{3x}{\sqrt{5x^2}} = \lim_{x\to-\infty} \frac{3x}{-x\sqrt{5}} = \frac{-3}{\sqrt{5}}$$

d. $k(x) = \dfrac{3^x-4}{5+3^x}$

HA: $y = 1$ $y = \frac{-4}{5}$

$$\lim_{x\to\infty} \frac{3^x-4}{5+3^x} = \lim_{x\to\infty} \frac{3^x}{3^x} = 1$$

$$\lim_{x\to-\infty} \frac{3^x-4}{5+3^x} = \frac{-4}{5}$$

$3^x \to 0$

as $x \to -\infty$

Example 15: Determine any asymptotes for the function $f(x)$ given the limit statements below.

$\lim_{x \to -\infty} f(x) = 4$ or $\lim_{x \to 3} f(x) = -\infty$

HA: $y=4$ VA: $x=3$